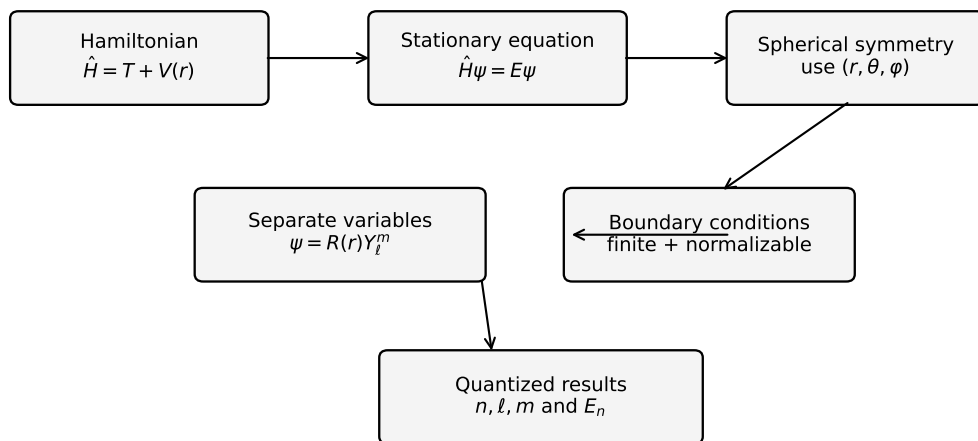


Solving the Schrodinger Equation for the Hydrogen Atom

Lecture notes for computer science students

How the hydrogen atom is solved



Goal. These notes explain how the hydrogen atom is solved in quantum mechanics: why energy levels are discrete, where the quantum numbers come from, how orbitals replace Bohr orbits, and how the Rydberg formula follows from the Schrodinger equation.

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1 From Bohr's atom to Schrodinger's atom

The Bohr model gives the correct energy levels for hydrogen:

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$

It also explains the main spectral formula:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad n_i > n_f.$$

But Bohr's model does this by adding a special postulate:

$$L = m_e v r = n \hbar.$$

This is useful, but it is not a full quantum theory.

Schrodinger's theory changes the question. Instead of asking for the electron's circular orbit, we ask for a wave function:

$$\psi(\mathbf{r}).$$

The wave function is not an orbit. It is a mathematical object from which probabilities are calculated.

Main idea

In the Schrodinger model, quantization is not imposed by hand as a special rule for circular orbits. Quantization appears because only some wave functions satisfy the physical boundary conditions.

What we want to obtain

By solving the Schrodinger equation for hydrogen, we want to obtain:

- the discrete energies

$$E_n = -\frac{13.6 \text{ eV}}{n^2};$$

- the quantum numbers

$$n, \quad \ell, \quad m;$$

- the orbital structure: $s, p, d, f, \dots$;
- the connection with spectral lines and the Rydberg formula.

For computer science intuition

Think of the hydrogen atom as an eigenvalue problem:

$$Av = \lambda v.$$

The Hamiltonian operator $\hat{H}$ plays the role of the matrix A . The wave functions ψ play the role of eigenvectors. The allowed energies E play the role of eigenvalues.

2 The stationary Schrodinger equation

The time-independent, or stationary, Schrodinger equation is

$$\hat{H}\psi = E\psi.$$

This says that an allowed stationary state is an eigenfunction of the Hamiltonian operator.

For a non-relativistic electron in a potential V , the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 + V.$$

Here μ is the reduced mass of the electron-proton system:

$$\mu = \frac{m_e m_p}{m_e + m_p} \approx m_e.$$

For most introductory calculations, μ can be replaced by m_e . Using μ is slightly more accurate.

2.1 Coulomb potential of the hydrogen atom

The electron is attracted by the proton. The potential energy is

$$V(r) = -\frac{k_e e^2}{r},$$

where

$$k_e = \frac{1}{4\pi\epsilon_0}.$$

Therefore, the stationary Schrodinger equation for hydrogen is

$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{k_e e^2}{r} \right] \psi = E\psi.$$

Important interpretation

The electron is not moving on a precise path around the nucleus. The function ψ gives probability information. The probability density is

$$|\psi(\mathbf{r})|^2.$$

3 Why spherical coordinates are natural

The potential depends only on distance from the nucleus:

$$V(r) = -\frac{k_e e^2}{r}.$$

It does not depend on direction. The hydrogen atom is therefore spherically symmetric.

So instead of Cartesian coordinates (x, y, z) , we use spherical coordinates:

$$(r, \theta, \varphi).$$

The Laplacian in spherical coordinates is

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}.$$

This expression looks unpleasant, but its structure is very helpful: there is a radial part and an angular part.

Symmetry principle

When the potential depends only on r , separate the wave function into radial and angular factors:

$$\psi(r, \theta, \varphi) = R(r)Y(\theta, \varphi).$$

4 Separation of variables

Because of spherical symmetry, we search for solutions in the form

$$\psi_{n\ell m}(r, \theta, \varphi) = R_{n\ell}(r)Y_{\ell}^m(\theta, \varphi).$$

This is the central structural result.

Part of wave function	Depends on	Meaning
$R_{n\ell}(r)$	distance r	radial distribution: how probability changes with distance from the nucleus
$Y_{\ell}^m(\theta, \varphi)$	direction	angular distribution: shape and orientation of the orbital

The angular functions Y_{ℓ}^m are called spherical harmonics. We do not need their full formulas in this lecture. We need to understand what they produce.

5 The angular equation and two quantum numbers

The angular part of the equation leads to angular momentum quantization. The allowed angular functions are labeled by two integers:

$$\ell = 0, 1, 2, \dots$$

and

$$m = -\ell, -\ell + 1, \dots, -1, 0, 1, \dots, \ell - 1, \ell.$$

These numbers have direct physical meaning.

Quantum number	Name	Meaning
ℓ	orbital angular momentum quantum number	orbital shape
m	magnetic quantum number	orbital orientation

The value of ℓ is connected with the magnitude of angular momentum:

$$L^2 = \ell(\ell + 1)\hbar^2.$$

The value of m is connected with one chosen component of angular momentum, usually the z -component:

$$L_z = m\hbar.$$

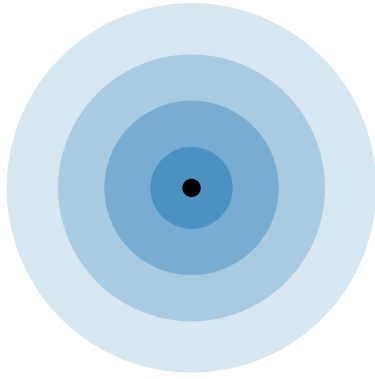
5.1 Orbital names

Chemistry and atomic physics use letters for ℓ :

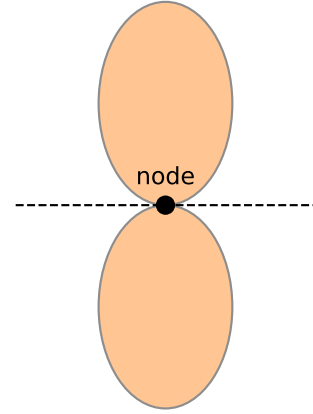
ℓ	0	1	2	3
orbital letter	s	p	d	f

Thus:

$$\ell = 0 \Rightarrow s, \quad \ell = 1 \Rightarrow p, \quad \ell = 2 \Rightarrow d, \quad \ell = 3 \Rightarrow f.$$



s orbital: spherical probability cloud



p orbital: two lobes and a nodal plane

Figure 1: Very schematic comparison of an s -type and a p -type orbital. These are probability clouds, not electron paths.

6 The radial equation

The radial part is where the energy levels appear.

After separation of variables, the radial function $R(r)$ satisfies

$$-\frac{\hbar^2}{2\mu} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{k_e e^2}{r} \right] R = ER.$$

This is the radial Schrodinger equation for hydrogen.

The term

$$\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2}$$

acts like an angular-momentum barrier. It is absent for $\ell = 0$, and it becomes stronger for larger ℓ .

A useful substitution is

$$u(r) = rR(r).$$

Then the radial equation becomes similar to a one-dimensional Schrodinger equation:

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{k_e e^2}{r} \right] u = Eu.$$

Effective potential

The radial motion behaves as if it sees an effective potential

$$V_{\text{eff}}(r) = \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{k_e e^2}{r}.$$

The first term is repulsive and comes from angular momentum. The second term is attractive and comes from the Coulomb force.

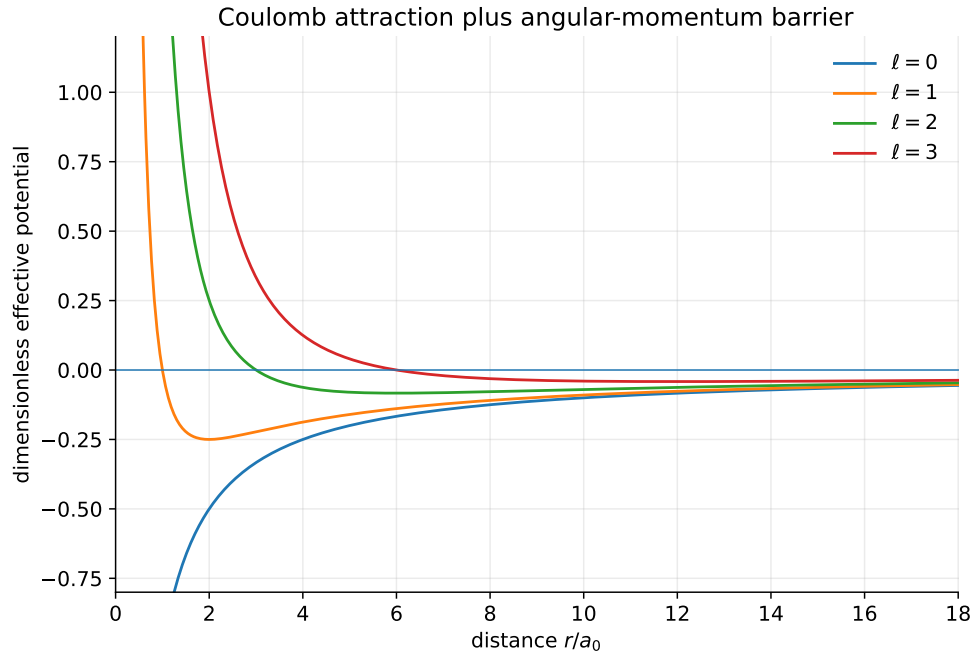


Figure 2: Dimensionless sketch of the effective potential. Larger ℓ creates a stronger barrier near the nucleus.

7 Boundary conditions: where quantization comes from

The radial equation has many mathematical solutions, but most of them are not physical.

A physically acceptable wave function must satisfy three requirements:

1. It must be finite near the nucleus.
2. It must vanish sufficiently fast as $r \rightarrow \infty$.
3. It must be normalizable:

$$\int |\psi|^2 dV = 1.$$

These conditions are the source of quantization.

Analogy with standing waves

A guitar string cannot vibrate with any wavelength. Only wavelengths that fit the boundary conditions are allowed. In hydrogen, the electron wave also has to “fit” the Coulomb potential and the boundary conditions. Only certain wave functions are allowed.

7.1 Asymptotic behavior

For bound states we have

$$E < 0.$$

Far away from the nucleus, the Coulomb potential becomes small and the wave function must decay exponentially:

$$u(r) \sim e^{-\kappa r}.$$

Near the origin, the physically acceptable behavior is

$$u(r) \sim r^{\ell+1}.$$

So the solution has the general structure

$$u(r) = \text{power near } 0 \times \text{exponential decay} \times \text{polynomial}.$$

More precisely, it has the form

$$u(r) \sim r^{\ell+1} e^{-r/(na_0)} \times \text{polynomial in } r.$$

The polynomial cannot be infinite in an arbitrary way. If it does not terminate, the solution eventually grows too fast and cannot be normalized. The requirement that the polynomial terminates produces an integer condition.

Quantization condition from the radial solution

The principal quantum number is

$$n = n_r + \ell + 1,$$

where $n_r = 0, 1, 2, \dots$ is the radial quantum number. Therefore

$$n = 1, 2, 3, \dots,$$

and for each n :

$$\ell = 0, 1, 2, \dots, n-1.$$

8 Energy levels

The allowed energies obtained from the radial equation are

$$E_n = -\frac{\mu k_e^2 e^4}{2\hbar^2} \frac{1}{n^2}.$$

Using the electron-proton reduced mass, this is approximately

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$

This is the same formula as in the Bohr model, but the interpretation is different.

n	Energy E_n	Meaning
1	-13.6 eV	ground state
2	-3.40 eV	first excited shell
3	-1.51 eV	second excited shell
4	-0.85 eV	higher excited shell
∞	0 eV	ionized electron

Meaning of negative energy

The zero of energy is chosen for a free electron infinitely far from the proton. A negative value means the electron is bound. To ionize hydrogen from the ground state, we must add 13.6 eV.

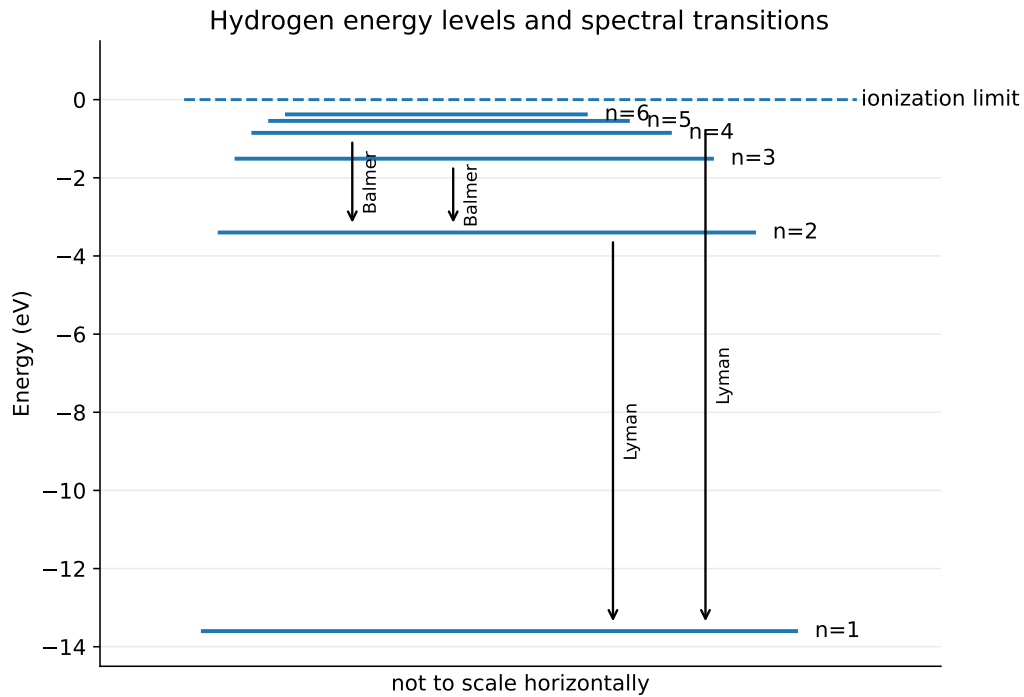


Figure 3: Hydrogen energy levels. Spectral lines correspond to transitions between these levels.

9 Allowed quantum numbers and orbitals

The complete stationary states of hydrogen are labeled by

$$\psi_{n\ell m}.$$

The allowed values are:

$$n = 1, 2, 3, \dots$$

$$\ell = 0, 1, 2, \dots, n-1$$

$$m = -\ell, -\ell+1, \dots, \ell-1, \ell.$$

9.1 Examples

For $n = 1$:

$$\ell = 0,$$

so there is only

$$1s.$$

For $n = 2$:

$$\ell = 0, 1,$$

so the orbitals are

$$2s, \quad 2p.$$

For $\ell = 1$, the magnetic quantum number can be

$$m = -1, 0, 1,$$

so there are three $2p$ orientations.

For $n = 3$:

$$\ell = 0, 1, 2,$$

so the orbitals are

$$3s, \quad 3p, \quad 3d.$$

9.2 Counting states without spin

For a fixed ℓ , there are

$$2\ell + 1$$

possible values of m .

For a fixed n , the total number of spatial states is

$$\sum_{\ell=0}^{n-1} (2\ell + 1) = n^2.$$

If electron spin is included, each spatial state can hold two spin states, so the number becomes

$$2n^2.$$

Degeneracy

In the ideal hydrogen atom, the energy depends only on n , not on ℓ or m :

$$E = E_n.$$

This means that different orbitals can have the same energy. This is called degeneracy.

10 The first wave functions

The ground state is the $1s$ state:

$$n = 1, \quad \ell = 0, \quad m = 0.$$

Its normalized wave function is

$$\psi_{100}(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}.$$

Here

$$a_0 = \frac{\hbar^2}{\mu k_e e^2}$$

is the Bohr radius.

Numerically,

$$a_0 \approx 5.29 \times 10^{-11} \text{ m}.$$

This is not a circular orbit

The maximum of $|\psi_{100}(r)|^2$ occurs at $r = 0$, but this does not mean the electron is simply located at the nucleus. To ask for the probability of a distance r , we must include the volume of a spherical shell.

10.1 Radial probability density

The probability of finding the electron between r and $r + dr$ is

$$P(r) dr.$$

For a spherically symmetric state,

$$P(r) = 4\pi r^2 |\psi(r)|^2.$$

For the 1s state,

$$P_{1s}(r) = 4\pi r^2 \left| \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0} \right|^2.$$

After simplification,

$$P_{1s}(r) = \frac{4r^2}{a_0^3} e^{-2r/a_0}.$$

This radial probability is maximal at

$$r = a_0.$$

Connection with Bohr

The most probable radius of the 1s electron in the Schrodinger model equals the Bohr radius a_0 . But the interpretation is different: it is the maximum of a probability distribution, not the radius of a classical orbit.

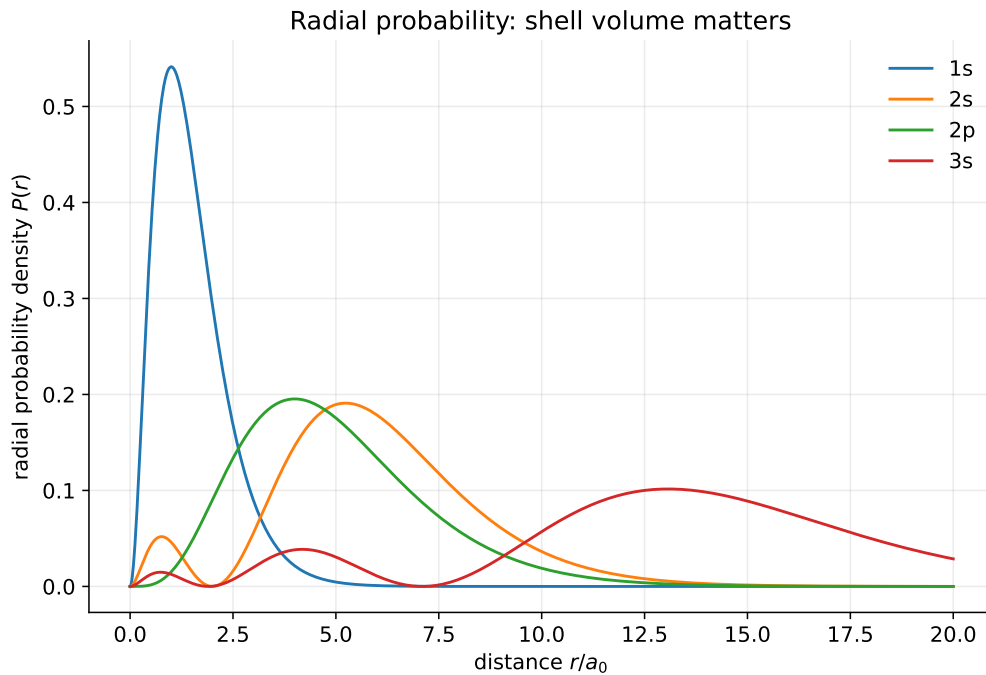


Figure 4: Radial probability distributions for selected hydrogen orbitals. Nodes and maxima appear naturally from the wave equation.

11 What the radial functions look like

A few low-energy hydrogen orbitals are useful to know.

11.1 The $1s$ orbital

$$\psi_{100}(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}.$$

Properties:

- spherical symmetry;
- no angular nodes;
- no radial nodes;
- lowest possible energy.

11.2 The $2s$ orbital

The $2s$ state has

$$n = 2, \quad \ell = 0, \quad m = 0.$$

It is still spherically symmetric, but it has one radial node. A simplified form is

$$\psi_{200}(r) \propto \left(2 - \frac{r}{a_0}\right) e^{-r/(2a_0)}.$$

The factor

$$2 - \frac{r}{a_0}$$

changes sign at

$$r = 2a_0.$$

That radius is a node of the wave function.

11.3 The $2p$ orbitals

The $2p$ states have

$$n = 2, \quad \ell = 1, \quad m = -1, 0, 1.$$

They are not spherical. They have angular structure. A typical real p -orbital has two lobes separated by a nodal plane.

Nodes

A node is a place where the wave function is zero. Nodes are not added by hand. They appear because the wave function must satisfy the Schrodinger equation and the boundary conditions.

12 Why energy depends only on n

In the ideal non-relativistic hydrogen atom, the energy formula is

$$E_n = -\frac{13.6 \text{ eV}}{n^2}.$$

It contains n , but not ℓ or m . Therefore, states such as

$$2s \quad \text{and} \quad 2p$$

have the same energy in the simplest theory.

This is special. In many other systems, energy depends on more quantum numbers. In real hydrogen, small corrections split some of these levels:

- spin-orbit coupling;
- relativistic corrections;
- Lamb shift;
- external fields, such as electric or magnetic fields.

For the main introductory picture, however, we keep

$$E = E_n.$$

13 From energy levels to spectral lines

When an electron changes from a higher energy level n_i to a lower energy level n_f , a photon is emitted.

The photon energy is

$$E_\gamma = E_i - E_f.$$

Using

$$E_n = -\frac{13.6 \text{ eV}}{n^2},$$

we get

$$E_\gamma = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

Since

$$E_\gamma = h\nu = \frac{hc}{\lambda},$$

we obtain

$$\frac{hc}{\lambda} = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

Therefore,

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad n_i > n_f.$$

This is the Rydberg formula.

13.1 Spectral series

Series	Final level n_f	Region
Lyman	1	ultraviolet
Balmer	2	visible / near ultraviolet
Paschen	3	infrared
Brackett	4	infrared

For the Balmer series,

$$n_f = 2,$$

so

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right), \quad n_i = 3, 4, 5, \dots$$

What Schrodinger explains

Balmer and Rydberg formulas work because the hydrogen atom has discrete energy eigenvalues. Spectral lines are transitions between eigenstates.

14 Worked example: the red Balmer line

The red hydrogen line H_α corresponds to the transition

$$n_i = 3 \longrightarrow n_f = 2.$$

The Rydberg formula gives

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{3^2} \right).$$

Compute the bracket:

$$\frac{1}{4} - \frac{1}{9} = \frac{9-4}{36} = \frac{5}{36}.$$

Thus

$$\frac{1}{\lambda} = R_H \frac{5}{36}.$$

Using

$$R_H \approx 1.097 \times 10^7 \text{ m}^{-1},$$

we get

$$\lambda \approx 6.56 \times 10^{-7} \text{ m}.$$

Therefore,

$$\boxed{\lambda \approx 656 \text{ nm}.}$$

This is visible red light.

15 Selection rules

Not every transition between energy levels is equally likely. For ordinary electric dipole radiation, the most important selection rules are

$$\boxed{\Delta\ell = \pm 1}$$

and

$$\boxed{\Delta m = 0, \pm 1.}$$

Examples:

$$2p \rightarrow 1s$$

is electric-dipole allowed because

$$\Delta\ell = 0 - 1 = -1.$$

But

$$2s \rightarrow 1s$$

is not an ordinary electric-dipole transition because

$$\Delta\ell = 0.$$

Practical lecture simplification

For deriving the Rydberg formula, we need only energy differences. For real spectral intensities and lifetimes, selection rules matter.

16 Bohr model versus Schrodinger model

Bohr model	Schrodinger model
Electron moves on circular orbits.	Electron is described by a wave function.
Quantization of angular momentum is postulated.	Quantization follows from boundary conditions.
One main quantum number n .	Three spatial quantum numbers n, ℓ, m .
Correct main hydrogen energies.	Correct energies plus orbital structure.
Semi-classical model.	Genuine wave-mechanical model.
Useful picture, but limited.	Foundation of atomic quantum mechanics.

One-sentence summary

Bohr guessed the correct energy pattern for hydrogen; Schrodinger explains that pattern as the spectrum of an eigenvalue problem.

17 Minimal mathematical derivation in one page

This section gives the shortest derivation path that is still logically complete.

Step 1: Start from the Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{k_e e^2}{r}.$$

The equation is

$$\hat{H}\psi = E\psi.$$

Step 2: Use spherical symmetry

Because $V = V(r)$, write

$$\psi(r, \theta, \varphi) = R(r)Y(\theta, \varphi).$$

The angular part gives

$$\ell = 0, 1, 2, \dots, \quad m = -\ell, \dots, \ell.$$

Step 3: Solve the radial equation

With $u = rR$:

$$-\frac{\hbar^2}{2\mu}u'' + \left[\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{k_e e^2}{r} \right] u = Eu.$$

Physical solutions must be finite and normalizable. This gives

$$n = n_r + \ell + 1, \quad n = 1, 2, 3, \dots$$

Step 4: Energies

$$E_n = -\frac{\mu k_e^2 e^4}{2\hbar^2} \frac{1}{n^2} \approx -\frac{13.6 \text{ eV}}{n^2}.$$

Step 5: Spectra

$$\frac{hc}{\lambda} = E_i - E_f = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

Therefore,

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

18 Lecture board plan

A good 60–90 minute lecture can follow this order:

1. Recall Bohr result: $E_n = -13.6 \text{ eV}/n^2$.
2. Explain what changes: orbits become wave functions.
3. Write the stationary Schrodinger equation $\hat{H}\psi = E\psi$.
4. Write the hydrogen Hamiltonian.
5. Explain spherical symmetry and separation of variables.
6. Introduce n, ℓ, m as labels of allowed states.
7. Show the radial equation and emphasize boundary conditions.
8. State the energy spectrum.
9. Explain orbitals and radial probability.
10. Derive the Rydberg formula from energy differences.

Recommended blackboard emphasis

Do not spend most of the lecture on special functions. Students should leave with the logical chain:

wave equation $\rightarrow$ boundary conditions $\rightarrow$ quantum numbers $\rightarrow$ energies $\rightarrow$ spectral lines.

19 Quick exercises with answers

Exercise 1

What are the allowed values of ℓ for $n = 4$?

Answer.

$$\ell = 0, 1, 2, 3.$$

So the possible orbital types are

$$4s, \quad 4p, \quad 4d, \quad 4f.$$

Exercise 2

How many spatial states exist for $n = 3$, ignoring spin?

Answer.

$$n^2 = 3^2 = 9.$$

Explicitly:

$$\ell = 0 \Rightarrow 1 \text{ state}, \quad \ell = 1 \Rightarrow 3 \text{ states}, \quad \ell = 2 \Rightarrow 5 \text{ states}.$$

Total:

$$1 + 3 + 5 = 9.$$

Exercise 3

What is the ionization energy from the level $n = 2$?

Answer.

$$E_2 = -\frac{13.6 \text{ eV}}{4} = -3.40 \text{ eV}.$$

To ionize from this level, the atom needs

$$3.40 \text{ eV}.$$

Exercise 4

Is the electric-dipole transition $3d \rightarrow 2p$ allowed?

Answer. For $3d$, $\ell = 2$. For $2p$, $\ell = 1$. Thus

$$\Delta\ell = 1 - 2 = -1.$$

Since $\Delta\ell = \pm 1$, this transition is allowed, assuming the magnetic quantum number also satisfies $\Delta m = 0, \pm 1$.

Exercise 5

What is the photon energy for the transition $n_i = 4 \rightarrow n_f = 2$?

Answer.

$$E_\gamma = 13.6 \text{ eV} \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = 13.6 \text{ eV} \left(\frac{1}{4} - \frac{1}{16} \right).$$

$$\frac{1}{4} - \frac{1}{16} = \frac{3}{16}.$$

So

$$E_\gamma = 13.6 \cdot \frac{3}{16} \text{ eV} \approx 2.55 \text{ eV}.$$

20 Final summary

What students should remember

1. Hydrogen is described by the stationary Schrodinger equation

$$\hat{H}\psi = E\psi.$$

2. The Coulomb potential is spherically symmetric:

$$V(r) = -\frac{k_e e^2}{r}.$$

3. Spherical symmetry lets us separate variables:

$$\psi_{n\ell m} = R_{n\ell}(r)Y_{\ell}^m(\theta, \varphi).$$

4. Boundary conditions produce quantum numbers:

$$n = 1, 2, 3, \dots, \quad \ell = 0, 1, \dots, n-1, \quad m = -\ell, \dots, \ell.$$

5. Energy levels are

$$E_n = -\frac{13.6 \text{ eV}}{n^2}.$$

6. Spectral lines come from transitions:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

The electron is not a planet.

It is a quantum state described by a wave function.